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Week 3 Quiz

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1. One of the problems with nearest neighbour resampling is (choose one of the following) **1 / 1 point**

- it can take a long time to compute compared with other resampling methods
- (•) it can produce a blocky (pixelated) final product if the output scale is very different from the scale of the recorded image data
- it requires more control points

Correct
That is correct

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2. When correcting geometric distortions using mapping polynomials a polynomial of high order (choose the correct answer below) **1 / 1 point**

- is always preferred
- (•) can lead to significant image distortions away from control points
- should never be used

Correct
That is correct. Good matches can be achieved at the control points but the behaviour can be poor in other parts of the image

3. When registering a set of images together using control points and mapping polynomial the preferred approach is (choose one from below) **1 / 1 point**

- to register image 2 to image 1, then image 3 to image 2, then image 4 to image 3, and so on
- (•) to register image 2 to image 1, then image 3 to image 1, the image 4 to image 1, and so on
- there is no preferred approach

Correct
That is correct, since registration errors cannot accumulate.

4. When using a second-degree mapping polynomial the minimum number of control points needed is (choose the correct answer) **1 / 1 point**

- 12
- 8
- (•) 6

Correct
That is correct

5. When choosing control points they must be (choose the correct answer) **1 / 1 point**

- spaced on a regular grid
- (•) distributed as evenly as possible over the image
- concentrated near regions of greatest brightness change in the image

Correct
That is correct. A well-spaced set of control points givens the best registration outcome.

6. If you suspect the distribution of control point is poor, for whatever reason, it is best to (choose one answer) **1 / 1 point**

- use a high degree mapping polynomial, such as a cubic
- (•) use a first degree mapping polynomial
- use any order (degree) since the degree of the polynomial is irrelevant

Correct
That is correct. This minimises geometric errors on the average, even though it might not be as good at individual control points as higher degree polynomials.

7. In general, would you expect a smoother looking geometrically corrected image if you (choose the correct answer) **1 / 1 point**

- nearest neighbour resampling
- bilinear interpolation resampling
- (•) cubic convolution resampling

Correct
That is correct. A smooth-looking resampled image will result.

8. When recorded, most remote sensing images are poor in contrast because (choose one answer) **1 / 1 point**

- (•) the detectors are designed to operate over a range that will detect absolute shadows and maximum brightness, such as clouds and snow
- generally, a radiometric resolution lower than desired is used for recording
- atmospheric absorption limits the brightness range that can be recorded.

Correct
That is correct.

9. When creating a colour image product from three recorded bands, it is conventional to assign the bands to the display primary colours of blue, green and red (choose the right answer) **1 / 1 point**

- (•) in increasing order of wavelength
- in decreasing order of wavelength
- in any order

Correct
That is the correct convention commonly used

10. Colour image products are important in photointerpretation because **1 / 1 point**

- They compress the information contents of the original bands
- (•) a human can generally perceive more detail in colour than in black and white
- it is easier to enhance contrast in colour than black and white

Correct
That is correct. It is a property of human vision systems.